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Single-Channel EEG-Based Seizure Prediction Using Deep Learning

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ABSTRACT

Reliable seizure prediction can improve patient safety by enabling timely protective actions, yet most high-performing approaches depend on multichannel EEG, limiting feasibility in wearable and low-power environments. This study developed an ultralight deep learning model for seizure prediction using only single-channel EEG and evaluated its clinical applicability under predefined, clinically actionable criteria: a seizure prediction horizon (SPH) of 2 minutes and a seizure occurrence period (SOP) of 30 minutes. A 37,985-parameter MobileNet-derived architecture was designed to process STFT spectrograms and validated using patient-specific leave-one-out cross-validation on the SNUH and CHB-MIT datasets. Performance was assessed using segment-based accuracy and false positive rate (FPR), along with event-based sensitivity computed under SPH–SOP constraints. The model demonstrated strong and consistent performance across both datasets. In the SNUH cohort, it achieved 85.97% accuracy, an FPR of 0.130, and 94.93% sensitivity, successfully predicting 95.08% of seizures within the SOP window. In the CHB-MIT dataset, it reached 90.72% accuracy, an FPR of 0.092, and 97.92% sensitivity. These findings provide the first evidence that single-channel EEG can support reliable seizure prediction within clinically meaningful early-warning windows. The proposed lightweight model delivers multichannel-comparable performance while drastically reducing computational demand, demonstrating strong potential for real-time deployment in wearable and patient-centered epilepsy management systems.

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1 Introduction

Epilepsy is a chronic neurological disease characterized by recurrent, unprovoked seizures, affecting approximately 50 million people worldwide [1]. Despite the development of new pharmacological and non-pharmacological treatments, including epilepsy surgery, vagus nerve stimulation, and deep brain stimulation, about one-third of patients remain refractory to current treatment modalities [2,3]. The unpredictable nature of seizure occurrence can impose a significant burden on patients' daily living and psychosocial well-being [4]. Seizure prediction technologies offer patients the ability to prepare in advance, reducing both physical and psychological risks and enhancing their quality of life [5]. Recent advances in artificial intelligence (AI) have significantly improved seizure prediction accuracy using EEG signals [7–9]. Models leveraging short-time Fourier transform (STFT) and attention mechanisms have achieved accuracies above 90%, effectively integrating local and global EEG features while capturing inter-channel dependencies. Transformer-based architectures using STFT images further enhance spatiotemporal pattern learning, reaching Area Under the Curve(AUC) values of up to 0.923 and sensitivities exceeding 93%. Additionally, Vision Transformer models employing Continuous Wavelet Transform (CWT) scalograms from multi-channel EEG data have demonstrated sensitivities and accuracies as high as 99.8% with false positive rates as low as 0.004. While highly effective, these methods often require intensive preprocessing and substantial computational resources, limiting their practical deployment in real-time or wearable settings. To improve computational efficiency, several studies have shifted toward simpler models using raw EEG data. LSTM-based architectures have been applied to capture temporal dynamics directly, while hybrid models combining CNNs for spatial features and LSTMs for temporal patterns have achieved high accuracy—up to 99.6%—with minimal preprocessing and low false positive rates [10]. Despite significant progress in seizure prediction, the majority of existing

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models rely on in-lab multichannel EEG data, which limits their applicability in real-world, ambulatory settings. This gap underscores the need for lightweight, computationally efficient models that can reliably predict seizures using single-channel EEG signals, particularly in the context of wearable devices. To address this critical gap, this study explores the feasibility of reliable seizure prediction using only a single EEG channel. We propose a lightweight deep learning model, adapted from the MobileNet architecture, specifically optimized for computational efficiency and minimal channel input. This paper details the model's development and evaluates its performance on two distinct scalp EEG datasets. Our goal is to establish a foundation for practical, long-term monitoring systems that balance predictive performance with the stringent constraints of real-world wearable applications.

2 Methods

2.1 . Dataset and predefined channel selection for single-channel evaluation

This study employed two EEG datasets—the Seoul National University Hospital (SNUH) dataset and the Children's Hospital Boston-Massachusetts Institute of Technology (CHB-MIT) dataset—for training and evaluating the seizure prediction model. The SNUH dataset includes recordings from 52 patients diagnosed with drug-resistant focal epilepsy, aged between 16 and 74 years. A total of 112 seizures were recorded using the international 10–20 electrode placement system at a sampling rate of 200 Hz. EEG signals were acquired from 21 scalp electrodes (Fp1, F3, C3, P3, Fp2, F4, C4, P4, F7, T1, T3, T5, O1, F8, T2, T4, T6, O2, Fz, Cz, and Pz), with a common average reference. The CHB-MIT dataset, a publicly accessible and widely cited benchmark in seizure prediction research, was also included for comparative analysis. It contains EEG recordings from 23 pediatric patients (aged 1.5 to 22 years) collected at Children's Hospital Boston. A total of 198 seizure events are annotated, and the data were recorded at a sampling rate of 256 Hz. EEG acquisition employed 18 bipolar channels: Fp1-F7, F7-T7, T7-P7, P7-O1, Fp1-F3, F3-C3, C3-P3, P3-O1, Fp2-F4, F4-C4, C4-P4, P4-O2, Fp2-F8, F8-T8, T8-P8, P8-O2, Fz-Cz, and Cz-Pz (Supplementary table 1). In this study, "single-channel EEG" denotes that the model receives one EEG derivation as the sole input at a time. To ensure feasibility for real-world deployment, we predefined a wearable-feasible candidate set prior to any model training. Specifically, we limited candidate channels to six scalp locations—Fp1, Fp2, T3/T7, T4/T8, O1, and O2—chosen to support easy attachment with minimal electrodes while maintaining coverage of common focal seizure onset regions. For the SNUH dataset, these six locations included as unipolar channels against common average reference. For the CHB-MIT dataset, which is provided in a bipolar montage, we selected bipolar derivations that best correspond to the same six scalp locations by mapping each predefined unipolar site to its closest available bipolar pair (e.g., Fp1–F7 and Fp2–F8 for frontal; T7–P7 and T8–P8 for temporal; P7–O1 and P8–O2 for occipital). Thus, the evaluated "single-channel" inputs represent a prespecified, cross-dataset-consistent channel set rather than a post hoc selection from all available channels. In addition, we assessed whether prediction performance differed by region (frontal/temporal/occipital) and laterality (left/right). As summarized in Table 2, we found no meaningful differences in prediction performance across regions and/or laterality, supporting the use of the predefined wearable-feasible channel set for single-channel evaluation. For model training and validation, only patients with at least two documented seizure events were included. Consequently, the final analysis comprised 62 seizures from 23 patients in the SNUH dataset and 54 seizures from 16 patients in the CHB-MIT dataset (Supplementary table 1). The seizure focus distribution for the SNUH final analysis set is summarized in Supplementary Table 2. In contrast, detailed seizure-type/focus annotations are not consistently available at the individual-event level in the CHB-MIT dataset, limiting stratified analyses by seizure subtype for that cohort. This study was approved by the Seoul National University Hospital Institutional Review Board (IRB No. H-2308-042-1456) and followed the principles of the Declaration of Helsinki. The requirement for written consent was waived due to the retrospective design.

2.2 Data Labelling

The entire EEG signals were categorized into three states—ictal, interictal, and preictal—relative to seizure onset [15]. This categorization enables the model to distinguish between normal brain activity, seizure events, and the critical preictal window where predictive features are likely to emerge. For both training and evaluation, only interictal and preictal data were used, excluding ictal data. The ictal state refers to the interval between the onset and the termination of a seizure. The onset and termination points were determined by two epilepsy specialists (JK and KT) through visual analysis of EEG signals and video recordings. In cases where the seizure boundaries were ambiguous, the two specialists reached a consensus to finalize the decision. Because the postictal state can persist for hours and differs from true interictal baseline [16], we conservatively defined a 3-hour postictal exclusion window. To prevent potential analytical bias from seizure clusters, any time segment between two seizures occurring within a 3-hour window was excluded from the analysis. Specifically, if a seizure began within 3 hours of the previous one's termination, the interval between them was not labeled as interictal or preictal and was omitted from both training and evaluation. The preictal state corresponds to an arbitrary interval before a seizure, where predictive information is presumed to exist. The definition of the preictal interval depends on two parameters: seizure prediction horizon (SPH) and seizure occurrence period (SOP) [15]. The SPH represents the minimum time before seizure onset during which a warning must

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occur. It ensures no seizures occur during this interval, and its validity is evaluated based on subsequent seizure occurrences. The SOP defines the time window following the SPH during which a warning is considered valid. A shorter SOP enhances the specificity of predicting seizure onset but risks missing seizures if set too narrowly. Thus, the SOP is determined by balancing prediction accuracy with practical applicability [15]. In this study, the SPH was set to 2 minutes, ensuring that warnings are issued at least 2 minutes before seizure onset, allowing patients to prepare. The SOP was set to 30 minutes, validating seizure occurrence within 30 minutes after the SPH. Consequently, the preictal state was defined as the interval from 2 to 32 minutes before seizure onset. A schematic overview of the EEG state labeling is shown in Supplementary Figure 1. The figure illustrates the temporal relationship among interictal, preictal, and ictal states, as well as the SPH, SOP, and the 3-hour exclusion windows applied around seizure onset.

2.3 Data Preprocessing

The sampling frequencies of all datasets were unified to the lowest sampling frequency by downsampling the CHB-MIT dataset to 200 Hz to ensure a consistent experimental environment while minimizing data loss. Additionally, we used a fifth order Butterworth band pass filter from 0.1 to 50 Hz to remove baseline drift and power noise. A sliding window approach was applied to both preictal and interictal data into 10-second windows with 50% overlap to maintain the continuity of signal characteristics while increasing the amount of training data [20]. The 10-second window length was selected to balance temporal resolution and spectral stability and has been commonly adopted in prior seizure prediction studies. Although some seizures in the CHB-MIT dataset have relatively short durations, the proposed framework focuses on identifying preictal patterns preceding seizure onset rather than modeling seizure duration itself, and is therefore applicable even for seizures with short ictal durations. For the interictal data, after excluding certain segments used for evaluation, random sampling was applied to the remaining interictal data for each epoch, ensuring that the total length of the sampled interictal data matched the length of the preictal data. This approach maximized the utilization of the dataset while maintaining a consistent ratio between the preictal and interictal data during training. The raw EEG signals were transformed into spectrograms in the time-frequency domain using the STFT. The STFT enables the analysis of the data, originally consisting of temporal information, as images containing both time and frequency components [21]. A Hann window with a frame length of 50 was used during the transformation process, with 50% overlap applied to enhance the time resolution. In the resulting spectrogram, frequencies corresponding to DC and those above 50 Hz were removed through slicing. As a result, the input data were represented as a tensor of size (80, 63, 1). The first dimension corresponds to the number of time frames generated by the sliding STFT window within a 10-second segment, the second dimension represents the number of frequency bins within the analyzed frequency range, and the third dimension denotes the single EEG channel used in this study.

2.4 Model design

The model used in this study is designed to operate on mobile devices and is based on the well-known MobileNet architecture. MobileNet utilizes a lightweight convolution operation called Depthwise Separable Convolution (DSC). DSC combines depthwise convolution and pointwise convolution to improve computational efficiency, enabling a lightweight model design while maintaining high performance [22]. When implementing layers with the same input and output using DSC, the computational load can be reduced by the number of output channels compared to conventional convolution. Additionally, stride (one of the hyperparameters of depthwise convolution) reduces the size of the feature map, thereby decreasing the computational load while preserving the essential characteristics. The low-resolution single-channel input data used in this study is not directly compatible with MobileNet. The repeated use of stride in deeper layers would significantly reduce the size of the feature maps, which may lead to potential loss of information. To mitigate this, we modified the original MobileNet architecture to maintain its lightweight computational efficiency while optimizing it specifically for low-resolution data, ensuring the preservation of essential features during training. While the original MobileNet contains 3,230,338 parameters, our modified architecture drastically reduces both the number of layers and channels, using only 37,985 parameters—just 1.18% of the original size (Supplementary Table 2). Furthermore, unlike MobileNet, the first convolutional layer in our model employs a stride of 1, minimizing information loss at the initial stage. The overall reduction in depth also helps prevent information degradation due to excessive feature map downsampling in deeper layers. This architecture is especially suitable for mobile devices, as it requires substantially less memory and can operate effectively in resource-constrained environments.

2.5 Patient-specific cross validation

Considering the inter-patient variability in seizure signal characteristics, patient-specific leave one out cross validation (LOOCV) was applied to ensure a reliable evaluation of the model's performance. The term "patient-specific" refers to constructing both the training and validation datasets only using the data from the same patient (i.e., wihtin-subject evaluation). Notably, "patient-specific" refers to within-subject modeling and does not imply a universal optimal channel across subjects or cohorts. In addition, candidate channels were restricted to a wearable-feasible set and evaluated accordingly (Fp1, Fp2, T3/T7, T4/T8, O1, O2). This approach allows the model to effectively learn and recognize the unique EEG characteristics and seizure patterns

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[23]. The final score for each patient was calculated by averaging their N trial results. Due to the temporal continuity of EEG signals, if data sampled from the same preictal segment is used in both training and evaluation, the model's reliability could be compromised [24]. To address this, each validation sample was constructed using one preictal segment determined by a specific ictal event during the labeling process, along with a 30-minute continuous segment randomly extracted from the interictal data. This setup ensures that the model is evaluated on entirely unseen preictal activity, thereby preserving the integrity of the assessment and its relevance to real-world seizure prediction tasks.

2.6 Postprocessing for evaluation

In consideration of the usability in clinical practice, the firing power method was applied to the individual window judgment results of the seizure prediction model [27]. This method was chosen for its ability to enhance the interpretability of the model's predictions and improve its practical application in real-world clinical settings. The firing power (FP) is calculated as follows.

$$FP[x] = \frac{\sum_{k=x-N}^{x} O[k]}{N} \tag{1}$$

$FP[x]$ represents the firing power value between 0 and 1 at time x, while N denotes the number of observed windows. $O[k]$ takes a value of 1 when it is more like preictal and 0 when it is more like interictal. The decision to generate an alarm based on $FP[x]$ is determined as follows.

$$a[x] = \begin{cases} \text{True}, & FP[x] \geq T, \\ \text{False}, & FP[x] < T, \end{cases} \tag{2}$$

Here, $a[x]$ indicates whether an alarm is generated, and T is a threshold value, which was set to 0.5 in this study. The selection of the smoothing window, $N = 119$, is empirically supported by performance metrics (Fig. 1) but more fundamentally reflects a deliberate clinical trade-off essential for real-world applicability [15]. A smaller window increases responsiveness to preictal signals but is highly susceptible to transient noise, elevating the false positive rate [15]. Frequent false alarms are a significant barrier to adoption, as they can induce patient anxiety and "alarm fatigue," ultimately disrupting daily life and eroding trust in the warning system [26]. Conversely, a larger window enhances system stability and reduces false alarms but introduces a critical risk of delaying the warning [15,26]. This delay could shorten the actionable window for intervention or cause brief, subtle preictal states to be overlooked entirely [15]. Therefore, the value $N = 119$ represents a carefully calibrated equilibrium, optimized to maintain high predictive sensitivity while minimizing the false alarm rate to a level deemed acceptable for long-term clinical use and patient quality of life [26]. Seizure alarms were triggered when the Firing Power value first exceeded the threshold of 0.5. All models were trained using the Adam optimizer with a learning rate of 0.001 and a weight decay of 0.0005. Binary cross-entropy was used as the loss function, and the evaluation metrics included binary accuracy, precision, recall, and the area under the ROC curve (AUC-ROC). A batch size of 16 was used for training, and models were trained for up to 300 epochs. To prevent overfitting, early stopping with a patience of 30 epochs was employed, starting from the 100th epoch. The best model was selected based on the highest validation accuracy.

2.7 Evaluation of seizure prediction performance

To quantitatively evaluate the seizure prediction model and verify its applicability in real-world settings, both Segment-based Evaluation and Event-based Evaluation metrics were employed. Specifically, Segment-based Evaluation metrics include Accuracy and FPR, while Event-based Evaluation focuses on Sensitivity [15]. Segment-based Evaluation assesses the model's ability to correctly predict each evaluation segment. In the case of binary classification tasks, performance metrics such as Accuracy and FPR are calculated using a confusion matrix. Accuracy represents the proportion of correctly predicted instances out of the total dataset, providing a measure of the model's overall performance. FPR, on the other hand, indicates the proportion of interictal segments misclassified as preictal, reflecting the occurrence of unnecessary warnings. A low FPR is essential for enhancing the reliability of the warning system, as it reduces false alarms and minimizes the burden on both patients and medical professionals. Event-based Evaluation, in contrast, focuses on determining whether the issued warnings accurately detect seizure events. Sensitivity is defined as the proportion of preictal segments determined by ictal events where warnings are issued within the desired period. This metric evaluates the model's success rate in predicting seizures in real-world scenarios.

2.8 Evaluation of single-channel performance in relation to seizure focus proximity

We evaluated whether single-channel prediction performance differed as a function of anatomical proximity between the EEG channel and the clinically determined seizure focus. For each patient, prediction performance was computed separately for six wearable-feasible scalp channels, yielding a repeated-measures structure (multiple channels nested within each patient). We

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operationalized "proximity to seizure focus" as a binary indicator (MatchFocus) based on concordance of both hemisphere and lobar region between the clinical focus label and channel location. Channels were mapped to hemisphere and region as follows: frontal (Fp1 left, Fp2 right), temporal (T3/T7 left, T4/T8 right), and occipital (O1 left, O2 right). Clinical focus labels were categorized by hemisphere (left, right, bilateral) and lobe (FLE, TLE, OLE). MatchFocus was coded as 1 when the channel was in the same lobe and ipsilateral to the focus; for bilateral foci, MatchFocus was coded as 1 for both hemispheres within the same lobe. All other channel–focus pairs were coded as 0. To test whether focus-matched channels improved accuracy (ACC), we fitted linear mixed-effects models with random intercepts for patient to account for within-patient correlation across channels: Model A: ACC MatchFocus + (1 | Patient). To control for systematic differences among channels independent of focus, we additionally fitted: Model B: ACC MatchFocus + Channel + (1 | Patient), where Channel was included as a categorical fixed effect. The incremental contribution of MatchFocus in Model B was assessed using a likelihood ratio test (LRT) comparing Model B to a nested model excluding MatchFocus. False-positive rate (FPR) was bounded (0–1) and exhibited substantial zero inflation (many observations with FPR = 0). Therefore, we used a two-part hurdle modeling strategy: Part 1 (occurrence): a mixed-effects logistic model for the probability of any false positives, Pr(FPR > 0), with fixed effects for MatchFocus and Channel and a patient-level random intercept. Part 2 (magnitude): among observations with FPR > 0, we modeled the magnitude of FPR after logit transformation, logit(FPR), using a linear mixed-effects model with MatchFocus and Channel as fixed effects and a patient-level random intercept. The incremental contribution of MatchFocus was tested using an LRT against a nested model excluding MatchFocus. Effects on the logit scale were exponentiated to report odds ratios for interpretability. Analyses were conducted in Python using mixed-effects modeling frameworks. Two-sided tests were used with $\alpha = 0.05$. Effect sizes are reported as regression coefficients with 95% confidence intervals (or approximate intervals for the logistic mixed model).

3 Results

The following sections detail the model's performance, evaluated using a subject-specific Leave-One-Out Cross-Validation (LOOCV) protocol. All reported performance metrics, including accuracy, FPR, and sensitivity, represent the average values calculated across all cross-validation folds for each patient.

3.1 Segment-based performance

A central finding of this study is that the model's predictive power was consistently highest when utilizing data from the prefrontal channels (Table 2). In the SNUH dataset, the Fp2 channel yielded the best performance, achieving an accuracy of 85.97% and a false positive rate (FPR) of 0.130 (Table 1). This finding was replicated in the CHB-MIT dataset, where the Fp1 channel demonstrated superior performance with 90.72% accuracy and an FPR of 0.092 (Table 3). Detailed segment-based performance results for the CHB-MIT dataset evaluated with $k = 60$ and $n = 119$ are provided in Supplementary Table 4, further supporting the robustness of these findings. This prefrontal advantage was further substantiated by regional analysis. Across both datasets, the frontal region consistently outperformed the temporal and occipital regions in all performance metrics. A comprehensive summary of channel-wise performance across all EEG channels, presented as mean ± standard deviation, is provided in Table 4. While this trend was robust, performance variability was observed at the individual patient level, with some patients demonstrating consistently high performance across all channels and others showing lower predictive accuracy. This variability underscores the patient-specific nature of seizure dynamics and highlights the importance of personalized modeling approaches.

3.2 Event-based performance

In the sensitivity analysis of seizure prediction shown in Table 3, the SNUH dataset shows the highest prediction success rate in the channel corresponding to Fp2 (94.93%). In contrast, the CHB-MIT dataset shows the highest prediction success rate in the channel corresponding to Fp1 (97.92%) (see Supplementary Table 4). However, certain channels failed to detect seizures for specific patients. For example, in the SNUH dataset, seizures were undetected in the T3 channel for patient 11, while in the CHB-MIT dataset, the P7-O1 channel failed to detect seizures for patient 11.

3.3 Seizure prediction pattern

Fig. 2 illustrates the seizure prediction patterns, showing the prediction value for individual windows (10 s), the timing of seizure alarms, and the actual seizure onset. This analysis highlights the model's ability to capture preictal states in most cases while acknowledging limitations in detecting some seizures and managing transient patterns. Out of 62 seizures in the SNUH dataset from the included SNUH patients ($\geq$2 seizures), 48 (77.4%) were successfully predicted with consistent preictal states detected before seizure onset. Among these, the most common pattern, observed in 43 seizures (Fig. 2a), involved detecting the preictal state starting 30 minutes before seizure onset. Additionally, 4 seizures (Fig. 2b) and 1 seizure (Fig. 2c) exhibited preictal states beginning 20 minutes and 10 minutes before onset, respectively. However, 11 seizures demonstrated

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less consistent or intermittent preictal patterns (Fig. 2d-f), and 3 seizures were not predicted by the model. In total, the model successfully predicted 95.16% of seizures, accounting for all cases in which preictal activity was present and could be reliably identified.

3.4 Seizure focus proximity effect on prediction performance

The analysis included 23 patients, each evaluated on six channels (total 138 patient–channel observations). In the unadjusted mixed model (Model A), MatchFocus was not associated with higher ACC ($\beta = -0.60$ percentage points; 95% CI -4.59 to 3.40; $p = 0.77$), indicating no detectable advantage for focus-matched channels in accuracy. After adjusting for channel identity (Model B), the MatchFocus effect remained nonsignificant and small ($\beta = 1.33$ percentage points; 95% CI -3.07 to 5.72; $p = 0.55$). The likelihood ratio test (LRT) comparing the adjusted models with versus without MatchFocus showed no incremental explanatory value of MatchFocus ($\chi^2(1) = 0.35$; $p = 0.55$).

Because the false positive rate (FPR) was frequently zero, we applied a hurdle mixed-model strategy. MatchFocus (Part 1) showed no clear association with the probability of observing any false positives (log-odds $= 0.19$; OR ≈ 1.21; approximate 95% interval 0.46 to 3.19). In contrast, conditional on FPR being nonzero (Part 2), focus-matched channels were associated with higher FPR. In the mixed-effects model on logit(FPR), MatchFocus was significantly positive ($\beta = 1.35$; SE $= 0.47$; $p = 0.004$), corresponding to an odds ratio of 3.84 for $\mathrm{FPR}/(1-\mathrm{FPR})$ (95% CI 1.52 to 9.73). The LRT confirmed improved model fit when adding MatchFocus ($\chi^2(1) = 7.29$; $p = 0.0069$).

4 Discussion

LightSeizureNet is a lightweight, interpretable deep learning model for real-time seizure detection on ultra-low-power devices[27]. While it demonstrates that compact architectures can achieve high detection accuracy. Our study builds on this lightweight approach but applies it to single-channel seizure prediction with validated SOP/SPH windows, enabling proactive rather than reactive clinical use. By setting the seizure prediction horizon (SPH) to 2 minutes and the seizure occurrence period (SOP) to 30 minutes, this study adopts a clinically meaningful configuration that balances early warning capability with patient usability. The SPH defines the minimum lead time before seizure onset during which a warning must be issued to be considered valid, while the SOP represents the window following the SPH within which a seizure is expected to occur. This approach is consistent with prior work by Truong et al. (2018), who used a 5-minute SPH and 30-minute SOP in their CNN-based prediction model[23], and with trends reviewed by Ren et al. (2022), which emphasize a 30-minute SOP as the most widely used standard due to its practicality and patient acceptance[15]. The 2-minute SPH in this study was carefully chosen to allow sufficient time for patients or caregivers to initiate precautionary actions before seizure onset, such as assuming a safe position, alerting assistance, or taking preemptive medication. Although shorter than the SPH used in some prior models, our design prioritizes minimal false predictions while maintaining high sensitivity. Ren et al. note that an SPH shorter than 2 minutes may limit intervention potential, while an overly long SOP could lead to unnecessary anxiety due to prolonged alert states. Importantly, these SPH and SOP values also support robust evaluation under event-based criteria. In this framework, a seizure is considered successfully predicted only if it occurs after the SPH and within the SOP, reducing the likelihood of misclassifying coincidental warnings as true positives. This rigorous setup enhances the clinical credibility of the model's predictions, making them both temporally precise and actionable in real-world, wearable-device scenarios.

Lu et al. (2024) also attempted single-channel EEG–based seizure prediction using STFT spectrograms and lightweight CNNs[13], but there are several key differences. While our study proposes channel selection optimized for wearable device applicability, Lu et al.'s work presented prediction results based on a randomly selected single channel. Such a method of channel selection is not suitable for wearable applications. Our study evaluated performance under predefined SOP (30 min) and SPH (2 min)—criteria essential for real-world applicability. We further validated a more compact model (37,985 vs. 213,597 parameters) on both a public dataset and a large in-hospital cohort, using patient-specific cross-validation and post-processing to minimize false alarms. These methodological advances yielded high sensitivity with low false positives, demonstrating stronger suitability for clinically reliable, wearable seizure prediction.

Across patients and channels, proximity to the clinically defined seizure focus (ipsilateral, same-lobe channel) did not confer a measurable improvement in prediction accuracy. This result remained unchanged after controlling for channel identity, suggesting that channel selection based solely on anatomical assumptions (e.g., selecting the electrode nearest the presumed focus) is not sufficient in a wearable-feasible scalp setting. This supports the concept that preictal dynamics detectable on scalp EEG are not strictly localized to the seizure onset zone, but can reflect distributed network-level changes and/or volume-conducted activity[24].

Although focus proximity was not associated with the probability of observing any false positives ($\Pr[\mathrm{FPR} > 0]$), it was associated with higher FPR magnitude conditional on FPR being nonzero. One possible explanation is that channels concordant with the clinical focus may capture more frequent preictal-like fluctuations or nonstationary activity patterns that increase false alarms, even if these do not improve overall discrimination enough to raise accuracy. This finding reinforces the need

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to jointly optimize accuracy and false-alarm burden when selecting a single wearable channel. Together, these results argue for patient-specific, data-driven channel selection rather than a priori focus-based selection. In practice, the "best" wearable channel should be chosen using an objective criterion that incorporates both predictive accuracy and false-alarm rate, aligning with the broader principle that personalization can reduce complexity while preserving performance[27].

Despite its advantages, the model exhibits certain limitations, particularly in cases where preictal segments are not consistently distinguishable. This highlights the necessity of further research to improve generalization across patients and datasets by addressing signal variability and dataset heterogeneity. Enhancing the model's robustness to inter-patient differences will be key to improving reliability in broader clinical contexts. To avoid contaminating the interictal class with postictal physiology that may persist for hours, we excluded intervals in which a new seizure began within 3 hours of the prior seizure termination. This conservative choice may underrepresent seizure-cluster contexts, which are clinically high-value periods where timely warnings could enable rescue interventions and help prevent further seizures within a cluster. Future work should explicitly address clusters (e.g., include a postictal/cluster state), report stratified performance for clustered versus isolated seizures, and evaluate sensitivity to different postictal exclusion windows.

Both datasets used for model development and evaluation predominantly comprise focal-onset seizures. The SNUH cohort includes only drug-resistant focal epilepsy cases, with seizure foci in the temporal, frontal, and occipital lobes. Therefore, our results support single-/few-channel seizure prediction primarily in focal epilepsy cohorts; performance in idiopathic/genetic generalized epilepsies was not evaluated and should not be inferred.

In addition, power efficiency is a critical requirement for mobile platforms[30]. Drawing on insights from energy-efficient mobile systems?such as intermittent communication, adaptive display configurations, and hardware-level optimizations?can further reduce power consumption without sacrificing performance [31]. These strategies will enable prolonged, real-world use of seizure prediction models by extending battery life while maintaining system stability and responsiveness [32].

5 Conclusion

This study highlights the potential of single-channel EEG data for reliable seizure prediction, emphasizing the importance of computational efficiency and real-time usability. By leveraging lightweight architecture and innovative post-processing techniques, our method bridges the gap between laboratory accuracy and clinical practicality. Future research should focus on expanding datasets, developing patient-specific generalization strategies, and refining energy-efficient deployment methods. These advancements will further establish single-channel EEG-based prediction as a feasible, patient-friendly solution for epilepsy management.

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Author Contributions

D.J. conceived and implemented the seizure prediction model, performed EEG preprocessing, training, evaluation, and led the manuscript writing. K.-Y.J. contributed to manuscript writing, supervised the overall clinical component, and provided clinical interpretation. Y.-G.J. organized and curated the EEG datasets. T.-J.K. provided clinical insights and assisted in medical interpretation. S.K.L. reviewed the clinical validity and approved the final version of the manuscript. K.-Y.M. contributed to manuscript revision and writing, led the system design and technical direction, and supervised the project from the engineering side. All authors reviewed and approved the final manuscript.

Ethics declarations

This study was approved by the Institutional Review Board of Seoul National University Hospital (IRB No. H-2411-051-1585). All procedures were conducted in accordance with relevant guidelines and regulations, including the Declaration of Helsinki and the Bioethics and Safety Act. Informed consent was obtained from all participants or their legal guardians.

Data availability

The CHB-MIT dataset is publicly accessible on PhysioNet. The SNUH EEG dataset is classified as protected clinical data and cannot be shared publicly. External transfer is permitted only with prior IRB approval and institutional authorization, and may be provided upon reasonable request to the corresponding author.

Competing interests

The authors declare no competing interests.

Supplementary information

Supplementary tables are provided in the supplementary file.

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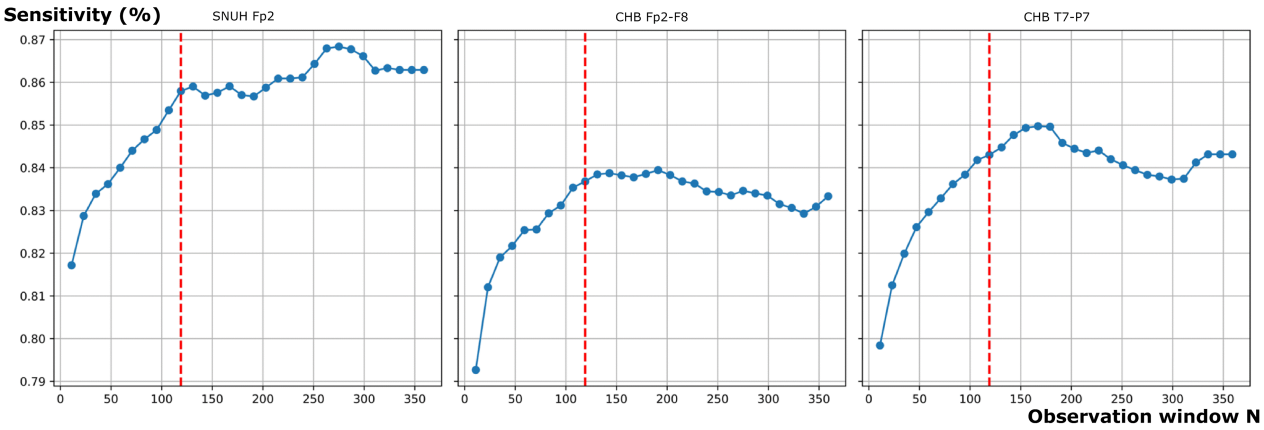

Figure 1. Representative segment-based accuracy changes as a function of varying *N* values in the firing power function for both the SNUH and CHB-MIT datasets. Larger *N* values result in smoother firing power curves, while smaller *N* values make the model more sensitive to short-term variations in preictal features.

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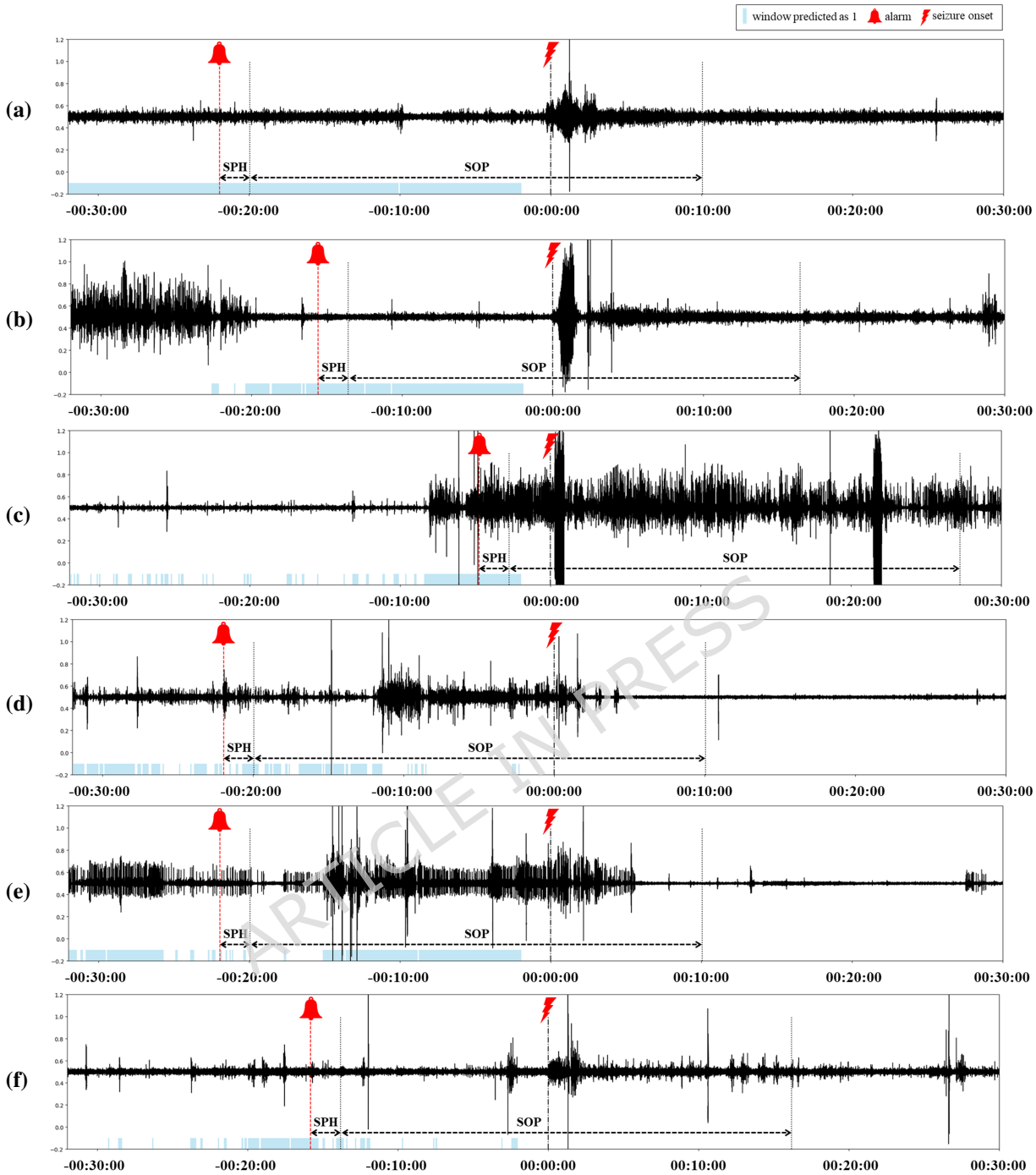

Figure 2. Representative seizure prediction patterns using the proposed model. Sky-blue regions indicate windows classified as preictal. (a) Consistent detection across the entire predictive window, with stable preictal characteristics observed consistently. (b) Detection beginning 20 minutes before seizure onset, with predictive features becoming apparent at this interval. (c) Detection occurring only when the seizure is imminent, shortly before onset. (d) Predictive characteristics appear consistently but disappear just before seizure onset. (e) Predictive features occur intermittently or in bursts within short intervals, without sustained detection. (f) Patterns characterized by transient or unstable prediction signals, showing less consistent preictal characteristics. Sky blue indicates individual window identified as the preictal state by our model.

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Table 1. **Segment-based evaluation in SNUH dataset with $k = 60$, $n = 119$**

Patient No.	Seizure focus	Fp1		Fp2		T3		T4		O1		O2	
		ACC	FPR	ACC	FPR	ACC	FPR	ACC	FPR	ACC	FPR	ACC	FPR
1	R TLE	100.0	0.000	100.0	0.000	97.29	0.000	94.69	0.000	96.25	0.000	95.52	0.000
2	R TLE	100.0	0.000	100.0	0.000	100.0	0.000	100.0	0.000	100.0	0.000	100.0	0.000
3	B TLE	89.00	0.167	89.06	0.167	91.67	0.167	90.07	0.167	84.44	0.097	89.10	0.000
4	L TLE	89.00	0.221	78.44	0.431	100.0	0.000	100.0	0.000	100.0	0.000	100.0	0.000
5	L TLE	93.65	0.000	79.17	0.107	87.03	0.114	69.64	0.570	65.57	0.590	50.10	0.250
6	B TLE	83.91	0.116	87.50	0.000	81.77	0.150	72.40	0.264	69.48	0.000	72.92	0.074
7	L TLE	75.00	0.500	100.0	0.000	50.00	0.000	75.00	0.500	75.00	0.500	74.48	0.000
8	L TLE	94.27	0.000	100.0	0.000	100.0	0.000	100.0	0.000	88.96	0.000	100.0	0.000
9	R TLE	100.0	0.000	77.85	0.000	92.92	0.094	96.32	0.000	91.32	0.000	95.76	0.000
10	R OLE	80.07	0.301	79.10	0.301	76.81	0.169	58.96	0.238	78.96	0.000	65.76	0.668
11	R FLE	51.04	0.979	50.31	0.994	61.35	0.273	44.38	0.160	63.44	0.000	77.60	0.448
12	L TLE	100.0	0.000	99.79	0.000	83.12	0.338	85.21	0.000	61.88	0.763	67.40	0.652
13	L FLE	100.0	0.000	83.44	0.000	100.0	0.000	75.00	0.000	100.0	0.000	100.0	0.000
14	L TLE	69.27	0.000	98.85	0.000	85.94	0.000	72.08	0.000	75.94	0.000	75.00	0.500
15	L TLE	100.0	0.000	98.96	0.000	89.86	0.000	100.0	0.000	94.51	0.000	98.68	0.024
16	R TLE	75.00	0.000	73.02	0.000	67.08	0.015	77.81	0.000	75.00	0.500	90.94	0.160
17	R TLE	89.79	0.204	89.03	0.219	70.28	0.000	91.88	0.163	99.72	0.006	88.82	0.224
18	L TLE	86.35	0.000	99.90	0.000	78.33	0.000	77.29	0.000	82.71	0.000	81.35	0.000
19	L OLE	74.58	0.081	72.60	0.000	76.46	0.075	73.23	0.000	82.81	0.344	71.15	0.002
20	R FLE	78.54	0.000	78.23	0.000	75.10	0.000	76.15	0.000	87.40	0.252	86.56	0.269
21	R TLE	76.28	0.151	82.53	0.142	69.58	0.382	78.30	0.138	63.65	0.212	79.51	0.195
22	L TLE	74.48	0.500	94.06	0.119	83.33	0.000	75.00	0.000	81.98	0.360	94.17	0.117
23	R TLE	84.17	0.000	65.52	0.500	71.25	0.031	58.96	0.231	65.21	0.000	70.42	0.310
AVG	–	85.41	0.140	85.97	0.130	82.14	0.079	80.10	0.106	81.92	0.158	83.71	0.169

Abbreviations: ACC, Accuracy; FPR, False Positive Rate; AVG, Average. R, right; L, left; FLE, frontal lobe. epilepsy; TLE, temporal lobe epilepsy; OLE, occipital lobe epilepsy

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Table 2. Segment-based performance by brain region

	SNUH		CHB-MIT	
Region	ACC	FPR	ACC	FPR
Frontal	85.78	0.134	86.94	0.121
Temporal	81.37	0.120	80.89	0.164
Occipital	81.55	0.163	84.65	0.105
Left	82.82	0.135	86.63	0.097
Right	82.98	0.143	81.69	0.165

Abbreviations: ACC, Accuracy; FPR, False Positive Rate.

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Table 3. **Event-based sensitivity in SNUH dataset**

Patient No.	Fp1	Fp2	T3	T4	O1	O2
1	100.0	100.0	100.0	100.0	100.0	100.0
2	100.0	100.0	100.0	100.0	100.0	100.0
3	100.0	100.0	100.0	100.0	83.33	83.33
4	100.0	100.0	100.0	100.0	100.0	100.0
5	100.0	75.00	100.0	100.0	100.0	50.00
6	100.0	75.00	100.0	75.00	50.00	75.00
7	100.0	100.0	0.000	100.0	100.0	50.00
8	100.0	100.0	100.0	100.0	100.0	100.0
9	100.0	100.0	100.0	100.0	100.0	100.0
10	100.0	100.0	100.0	66.67	66.67	100.0
11	100.0	100.0	50.00	50.00	50.00	100.0
12	100.0	100.0	100.0	100.0	100.0	100.0
13	100.0	100.0	100.0	50.00	100.0	100.0
14	50.00	100.0	100.0	50.00	100.0	100.0
15	100.0	100.0	100.0	100.0	100.0	100.0
16	50.00	50.00	50.00	100.0	100.0	100.0
17	100.0	100.0	66.67	100.0	100.0	100.0
18	100.0	100.0	100.0	100.0	100.0	100.0
19	100.0	100.0	100.0	100.0	100.0	50.00
20	100.0	100.0	100.0	100.0	100.0	100.0
21	83.33	83.33	83.33	83.33	66.67	83.33
22	100.0	100.0	100.0	50.00	100.0	100.0
23	100.0	100.0	100.0	100.0	50.00	100.0
AVG	94.93	94.93	89.13	88.04	89.86	90.94

Abbreviations: AVG, Average.

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Table 4. **Performance metrics of the proposed model across different EEG channels. All values are presented as mean ± standard deviation.**

Dataset	Channel	Accuracy	Precision	Recall	F1-score	FPR
CHB-MIT	FP1–F7	0.902 ± 0.155	0.919 ± 0.189	0.903 ± 0.220	0.892 ± 0.191	0.099 ± 0.253
	FP2–F8	0.837 ± 0.196	0.845 ± 0.298	0.811 ± 0.332	0.793 ± 0.298	0.137 ± 0.310
	T7–P7	0.843 ± 0.202	0.802 ± 0.353	0.791 ± 0.366	0.776 ± 0.344	0.105 ± 0.272
	T8–P8	0.775 ± 0.226	0.725 ± 0.371	0.771 ± 0.395	0.711 ± 0.362	0.220 ± 0.388
	P7–O1	0.854 ± 0.203	0.831 ± 0.333	0.789 ± 0.375	0.780 ± 0.350	0.081 ± 0.233
	P8–O2	0.838 ± 0.213	0.801 ± 0.344	0.804 ± 0.362	0.780 ± 0.340	0.127 ± 0.283
SNUH	Fp1-AVG	0.857 ± 0.182	0.868 ± 0.258	0.851 ± 0.268	0.837 ± 0.247	0.136 ± 0.302
	Fp2-AVG	0.858 ± 0.189	0.863 ± 0.279	0.844 ± 0.298	0.827 ± 0.275	0.128 ± 0.303
	T3-AVG	0.821 ± 0.187	0.834 ± 0.308	0.749 ± 0.367	0.745 ± 0.331	0.106 ± 0.235
	T4-AVG	0.805 ± 0.216	0.802 ± 0.340	0.741 ± 0.397	0.720 ± 0.369	0.131 ± 0.292
	O1-AVG	0.806 ± 0.193	0.779 ± 0.347	0.767 ± 0.347	0.748 ± 0.321	0.156 ± 0.311
	O2-AVG	0.826 ± 0.196	0.784 ± 0.326	0.815 ± 0.344	0.775 ± 0.327	0.163 ± 0.293

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